# 01

# Code specifications

12/09/2025

# Cuts applied in Matteo's original code (per channel): MuTau

| mutau | | | | |
|---|---|---|---|---|
| | ttjets | dy | qcd | data |
| FASE0 | 1 tau. 1 muon | 1 tau. 1 muon | 1 tau. 1 muon | 1 tau. 1 muon |
| Doube_medium | | | | |
| Iso_Mu24 | ✓ | ✓ | ✓ | ✓ |
| Ele32_WpTight | | | | |
| Electron_ | | | | |
| lumi filtering | | | ✓ | ✓ |
| | | | | |
| FASE1 | | | | |
| tau id1 | >63 | >63 | >63 | >63 |
| tau id2 | >7 | >7 | >7 | >7 |
| tau id3 | >1 | >1 | >1 | >1 |
| tau pt | >100 | >100 | >100 | >100 |
| mu id | >=3 | >=3 | >=3 | >=3 |
| mu pt | >35 | >35 | >35 | >35 |
| ele id | | | | |
| ele pt | | | | |
| delta R | >0.4 | >0.4 | >0.4 | >0.4 |
| eta | <2.4 | <2.4 | <2.4 | <2.4 |
| charge signs | opposite | opposite | same | opposite |
| scaling factors | ✓ | ✓ | | |
| valid jet and MET info | ✓ | ✓ | | |

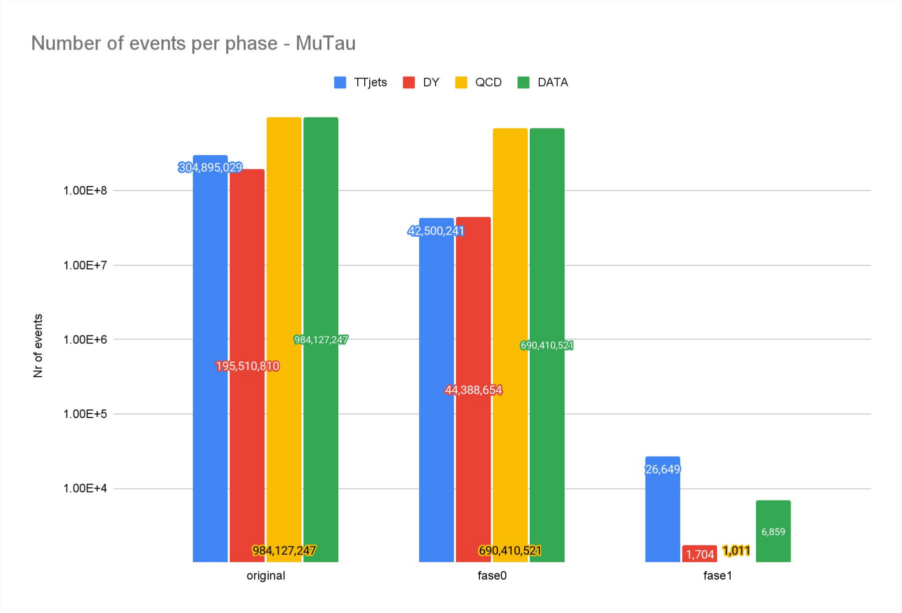
| Selection cuts |
|---|
| 1 muon and 1 $\tau_h$ (ID+ISO) |
| $p_T(\mu) > 35\,\mathrm{GeV}$, $p_T(\tau_h) > 100\,\mathrm{GeV}$ |
| Opposite sign (same-sign for QCD) |
| At least two protons in PPS |

Cuts explained in the analysis note

proton requirements are applied after introducing pileup protons

# MuTau Channel - comparison in number of events

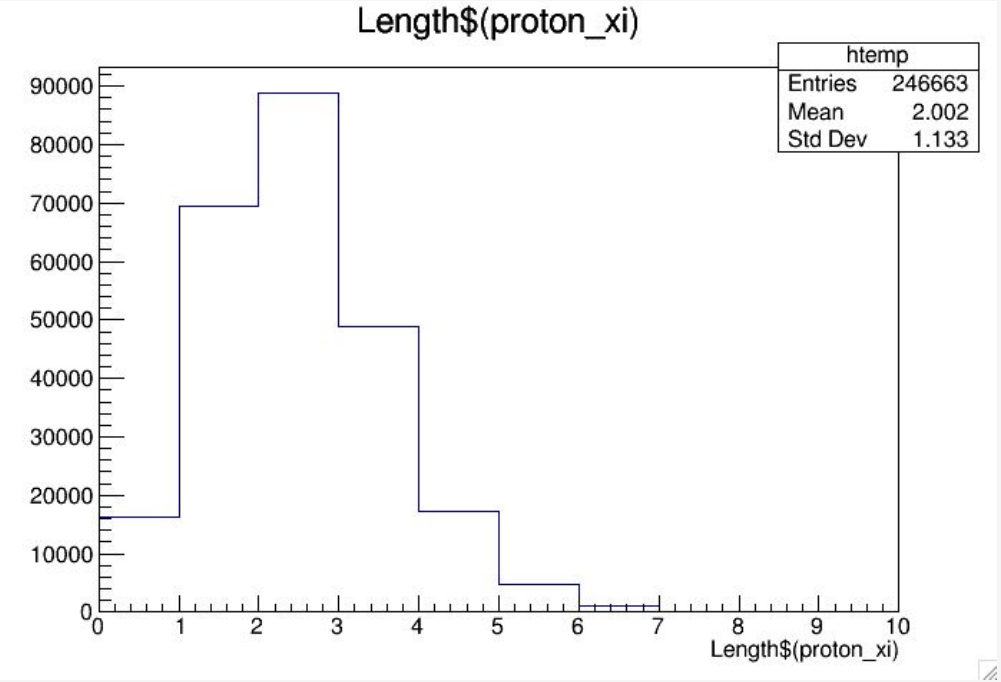
| | | original samples | my code | original samples | my code | SingleMuon | | | |
|---|---|---|---|---|---|---|---|---|---|
| | original | 304,895,029 | | 195,510,810 | | 984,127,247 | | | |
| MuTau | fase0 | 15,681,844 | 42,500,241 | 39,736,913 | 44,388,654 | 765,341 | 690,410,521 | 299,979 | 690,410,521 |
| | fase1 | NA | 26,649 | NA | 1704 | NA | 1011 | NA | 6859 |
| | fase1+pileup protons | NA | | NA | | NA | | NA | |



# BEFORE signal processing: starting from

MINIAOD data

Tails indicate pileup protons are already included in this sample (before the sinal.cpp code)



# signal processing: starting from MINIAOD data

## 2 Signal and background samples

This study considers the 2018 data taking period. For this reason, samples are simulated taking in account the behaviour of the detector configuration during this year.

### 2.1 Signal samples

In this analysis, we are interested in testing the SM predictions and BSM physics models. The signal sample is generated using MadGraph5_aMC@NLO v3.5.2 [8–10], which includes recent charge form factors of photon fluxes and hadronic survival probabilities [11]. Two BSM models are considered: high mass resonances and EFT modifications of the $\gamma\tau$ coupling. The BSM model[1] is implemented using the SMEFTsim package [12], with $\alpha^{-1} = 137$.

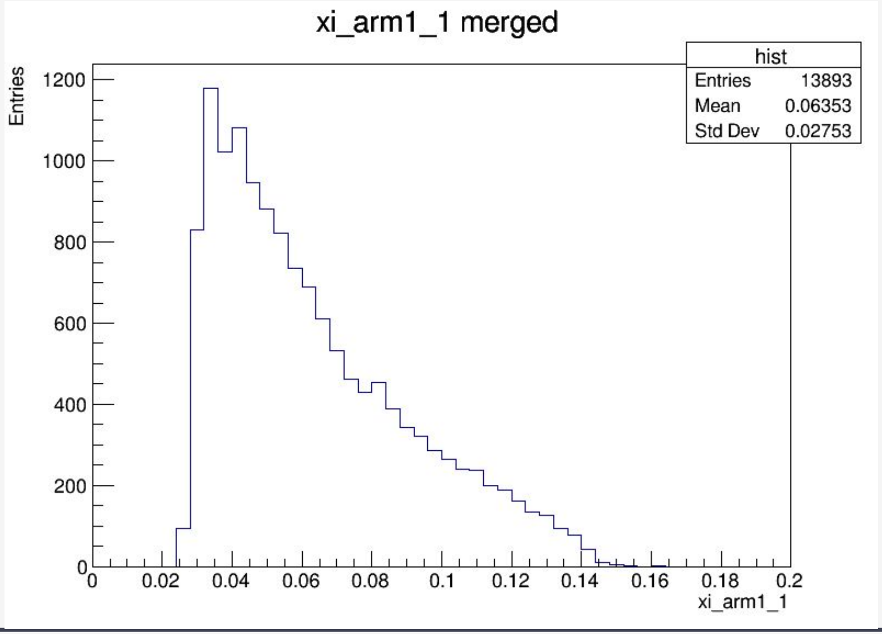
- Reads muon scale factors
- loads proton acceptance correction and systematic uncertainties
- Muon corrections: Trigger, Id, ISO, Reco
- Applies tau/muon ID cuts + η
- Normalization with BSM weights in proper branches

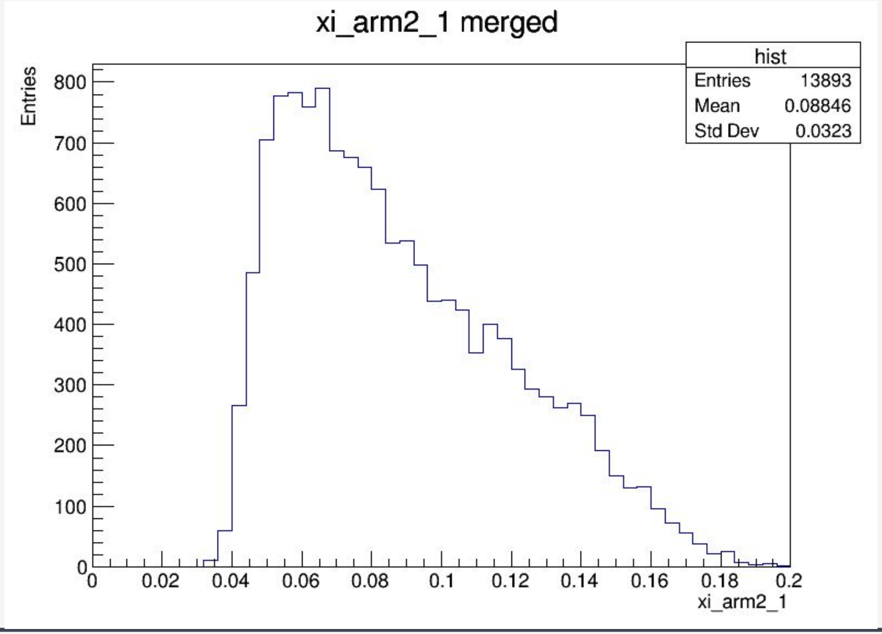
(proton cuts expanded in next slide)

# Pronton ξ AFTER sinal.cpp

Proton related cuts:

- 2 reconstructed protons
- fiducial acceptance cuts - base on ξ and θx
- period dependent geometric cuts
- radiation damage corrections - weight factors applied from proper root files according to run period
- ξ systematic variations - evaluation of systematic uncertainties -> produces up and down shifted ξ values per arm



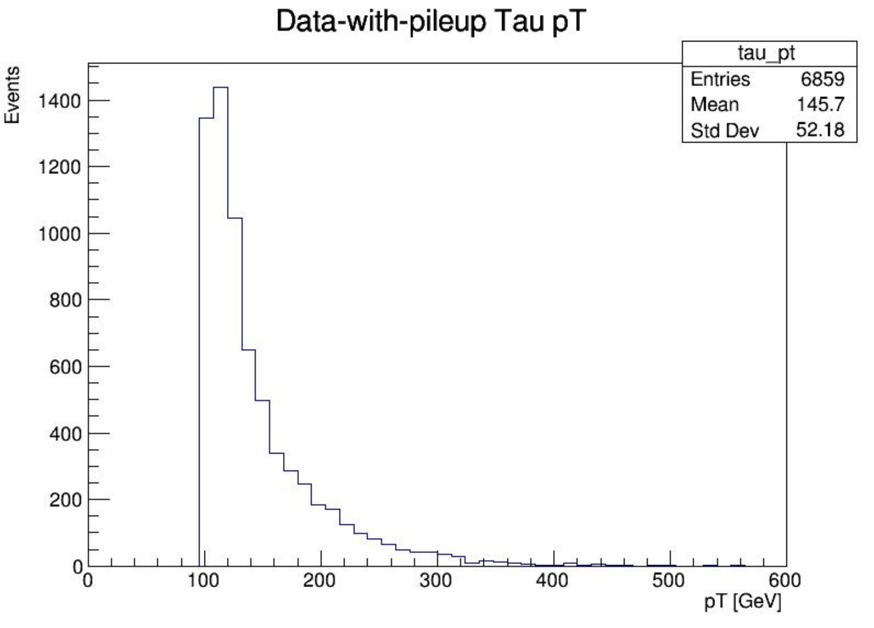



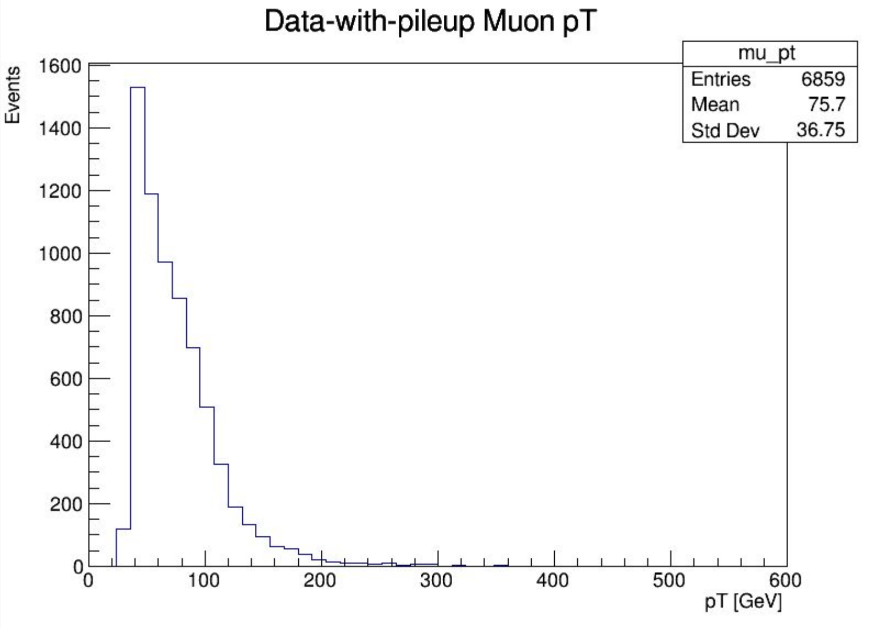
# 02

# Variable plots

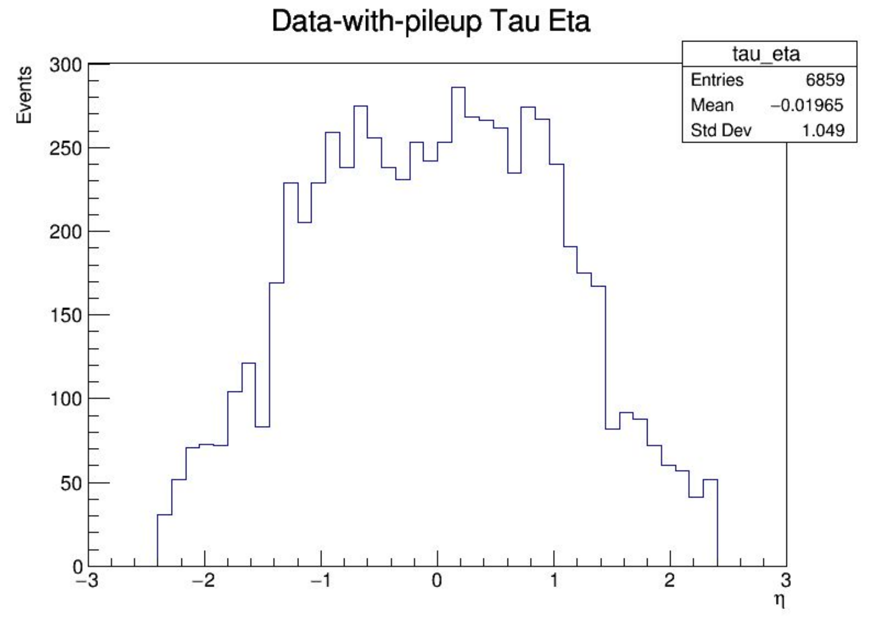
Data

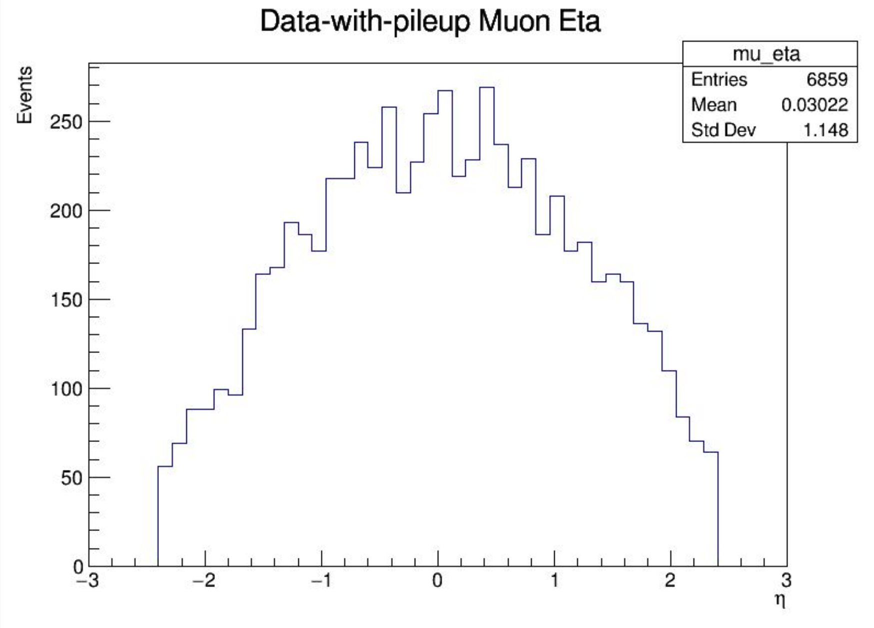
# Plots after adding pileup protons -> Data final samples



# Plots after adding pileup protons
# -> Data final samples



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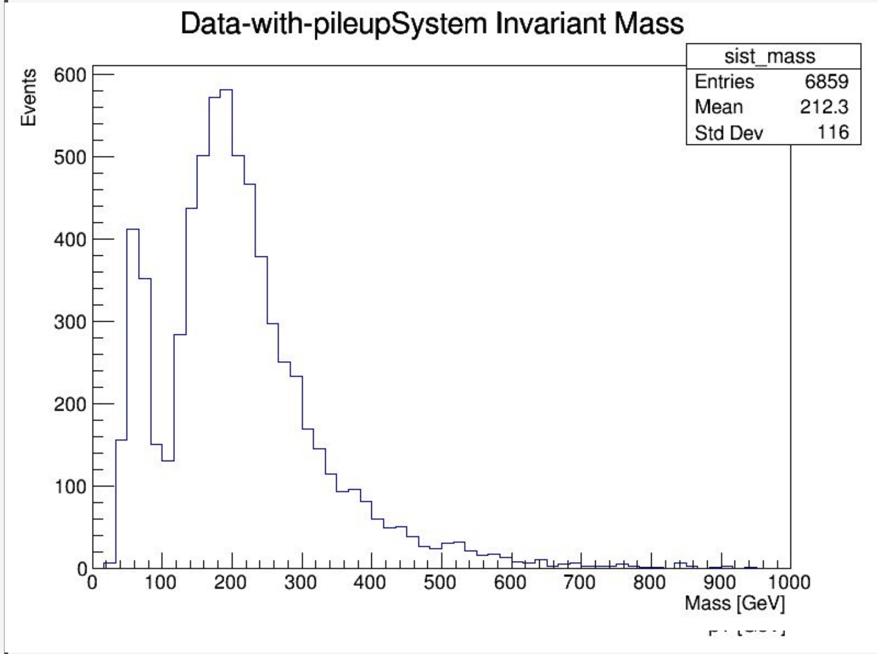
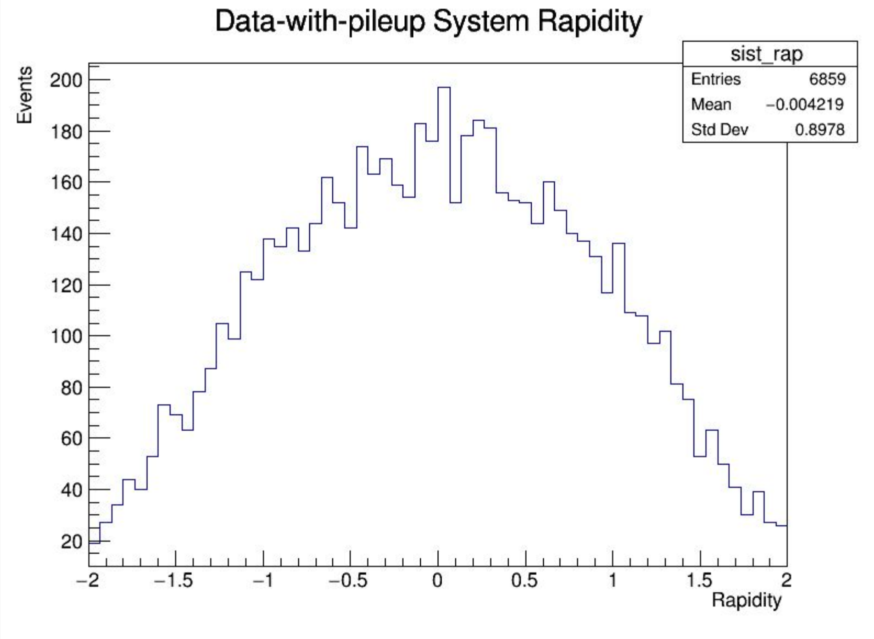
# -> Data final samples



# 02

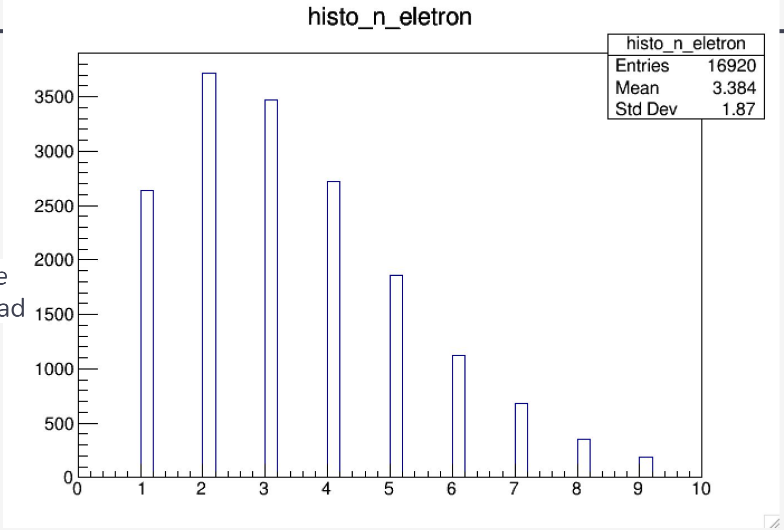
# Variable plots

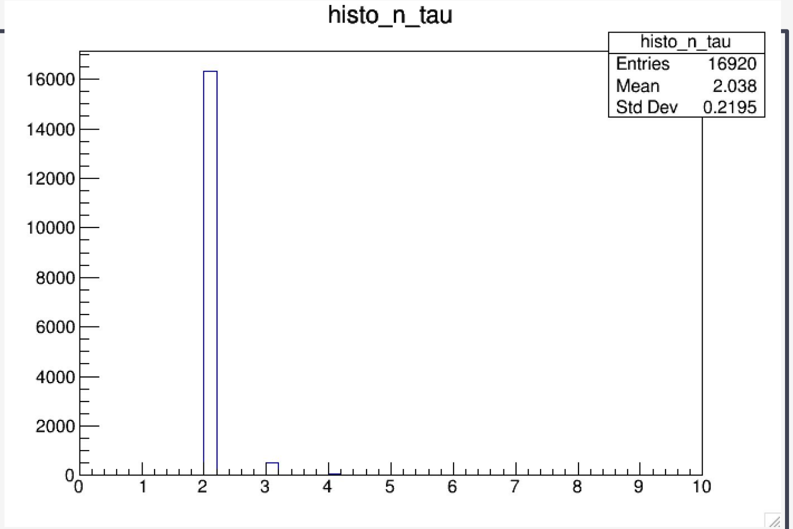
Signal

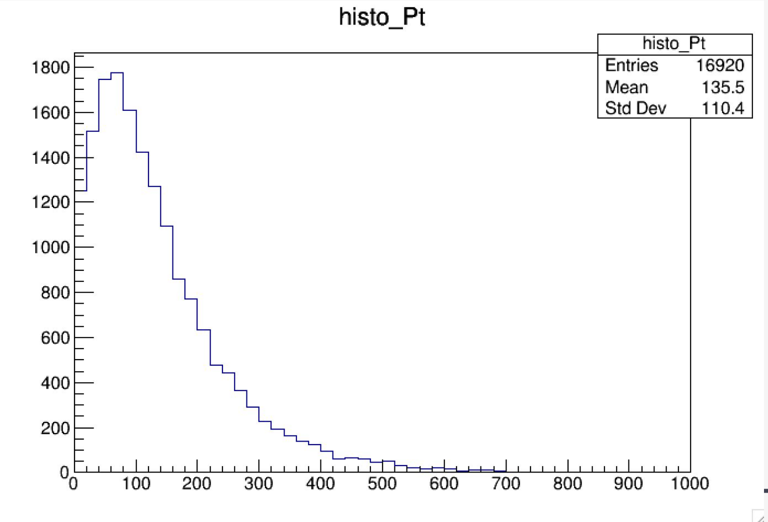
## Signal plots produced by sinal.cpp code mutau channel

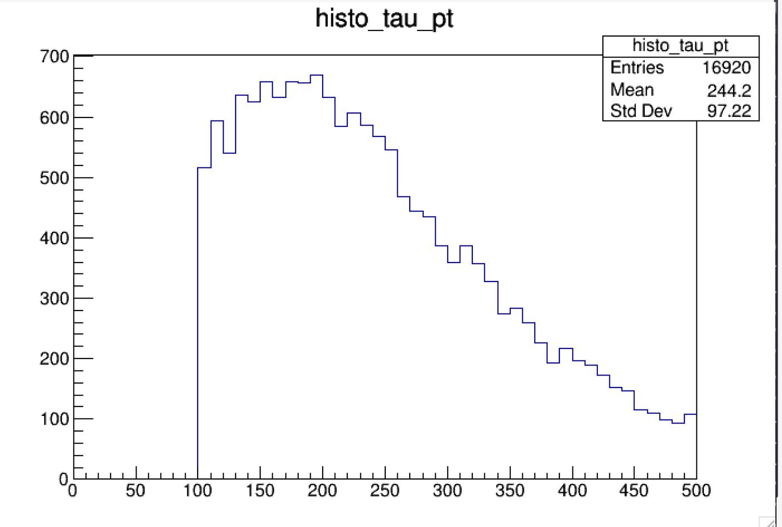
note: where is written e or *electron* it should read *muon*

- mu pT>35
- tau pT>100
- eta<2.4 for both
-
- opposite sign candidates
-
- at least one proton per pps arm



# Signal plots produced by sinal.cpp code mutau channel

note: where is written e or *electron* it should read *muon*

- mu pT>35
- tau pT>100
- eta<2.4 for both
- 
- opposite sign candidates
- 
- at least one proton per pps arm